Fine-Grained Behavioral Fingerprints for Terminal Traces

A Reusable, Interpretable Measurement Layer for Independent Analytic Tasks

AttackAgent Detect

12 November 2026

Abstract

We define 22 interpretable B/S/T statistics for local action organization, sequence evolution, and observed launch timing in terminal traces. Our primary evaluation asks whether pairwise fingerprint distances improve same-task, same-mode retrieval on CTH and AIT; we also study cross-task configuration retrieval on an Agent-only NYU cohort. As an auxiliary application, category-normalized B/S/T produces a single-log Agent-direction score on AIT. This score is never an input to association. Across these previously inspected cohorts, B/S(/T) raises Hit@1 point estimates but also false acceptance. The auxiliary A/H readout reaches 75.8% task-held-out and 59.6% source-held-out balanced accuracy, below command-identity references. The results support reuse of interpretable measurements across analytic tasks, not universal transfer, actor identity, or APT attribution.

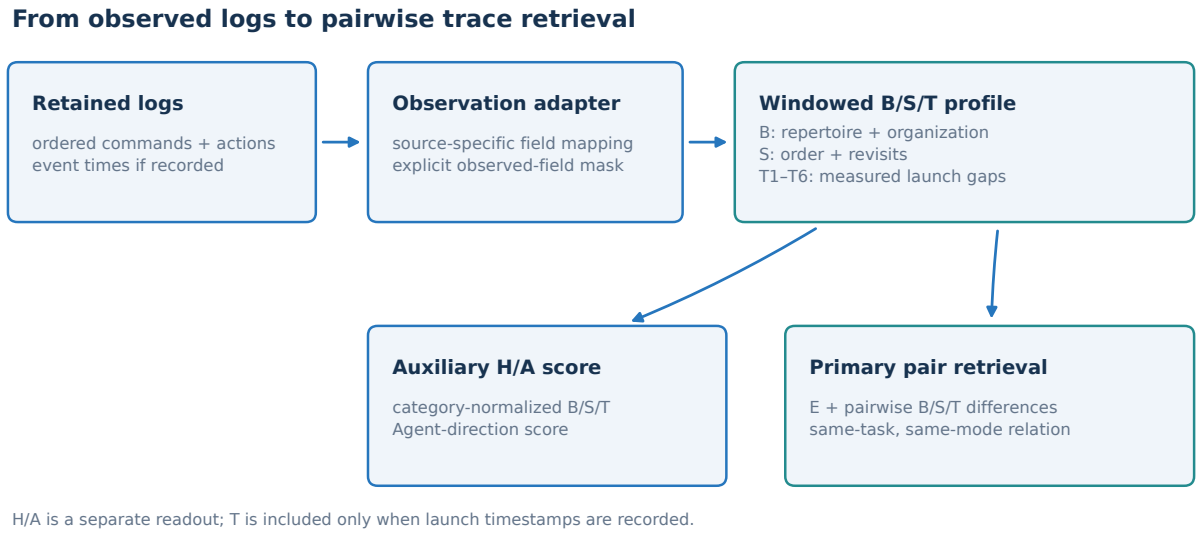


Figure 1: Three-stage path from retained logs through source-aware observation mapping to a windowed B/S/T profile. Pair retrieval is the primary application; the auxiliary Human/Agent score is not an association input. T is included only when launch timestamps are recorded.

1 Introduction

Incident analysts often receive command records that reveal tools and actions but little provenance. Two traces may use the same executable while differing in how actions are organized, ordered, or spaced in time. We ask whether these observable behaviors help retrieve traces from the same benchmark task and recorded execution mode, beyond command and action n-gram similarity.

We define a task-independent fingerprint of 22 named B/S/T measurements. A pairwise model compares traces using command evidence and their behavioral profiles. As a separate auxiliary application, category-normalized B/S/T can produce a continuous Agent-direction score for one log; that score never enters the association model. Here, reuse means applying the same interpretable measurements to different analytic tasks, while direct model transfer across sources is evaluated separately.

An earlier CTH study provides background for the threshold problem. Figure 2 varies the Agent share in constructed galleries while using a fixed command or action-bigram threshold. Its same-task-only label treats same-task cross-mode pairs as positive, unlike the same-task, same-mode relation studied here. The curve motivates the new evaluation; it is not evidence that B/S/T improves the new target.

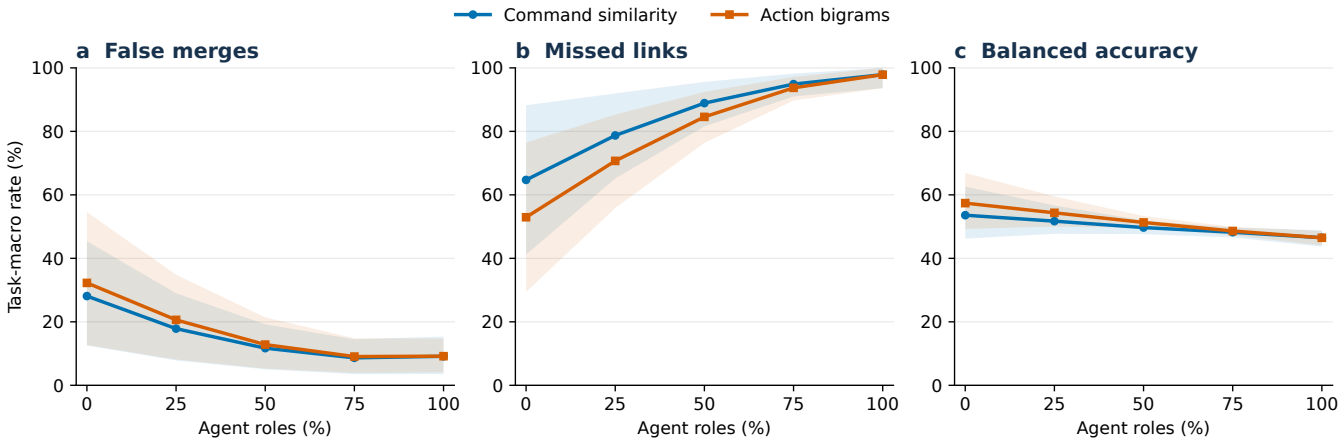


Figure 2: Command and action-bigram baselines across five CTH gallery role shares under the historical same-task-only label. Panels show false merges, missed links, and balanced accuracy at a fixed threshold; shaded bands indicate 95% task-cluster intervals.

We evaluate association on CTH (17 tasks, 85 traces), AIT (12 tasks, 109 traces), and an Agent-only NYU D-CIPHER cohort. We separately evaluate the single-log A/H use case on the existing AIT labels. Our contributions are: (1) a fine-grained, interpretable B/S/T measurement schema for terminal traces; (2) two independent instantiations of its feature meanings, for pairwise comparison and single-log A/H review; and (3) held-out results that report each application's performance and limits. The AIT data are troubleshooting logs, and all conclusions remain bounded by the evaluated sources and labels.

2 Related work

2.1 Behavioral identity and workflow

Trace extracts model-family signals from terminal commands, LIDAR distinguishes coding models with active probes, Known By Their Actions studies browser-model signatures, and ProcGrep represents coding-agent traces as workflows [2, 7, 8, 6]. HABITS uses grouped session-level behavior signals to separate Human and shopping-Agent sessions, while showing that timing-only evidence weakens when timing is imitated or matched [3]. These studies illustrate distinct uses of behavioral measurements, while their identity and shopping labels do not establish trace association in security logs. Our focus is a shared, inspectable vocabulary for host-terminal traces: the same B/S/T measurement meanings can be read at the trace level and instantiated in pairwise retrieval or single-log review. We evaluate these objectives separately, with application-specific labels and outputs.

Earlier UNIX user-profiling and network-security user models also examined command order and execution timing [14, 13]. McMahon and Sholl found their command-string metric more promising than the timing metric, which they said needed refinement [13]. This history motivates testing observed sequence and timing evidence, while leaving the value of any particular feature to matched evaluation.

2.2 Cyber-agent behavior and security labels

CyBiasBench analyzes Agent behavior from HTTP request traces labeled by Agent, target, and attack family [9]. It supports Agent-side preference analysis, but its public cohort has no Human traces and its request-level view differs from host-command logs. HABITS directly studies Human/Agent separability in browser-shopping sessions, with an explicitly different observation interface and scenario [3]. Synthetic APTs studies controlled Agent personas and warns against turning emulated behavioral resemblance into actor attribution [1]. We use these works to motivate two distinct fingerprint applications while keeping their targets separate.

2.3 Attack correlation and relation evidence

Attack-correlation systems align temporal, causal, provenance, or technique evidence to reconstruct or search for attack behavior [17, 19, 16, 18, 20]. Cross-Agent links asynchronous sessions using proxy-visible events, text, timing, and synthetic campaign labels [10]. These systems provide richer activity or campaign evidence than the benchmark-task labels available in our cohorts. We therefore treat a shared task only as a retrieval proxy: it does not establish shared attack activity, campaign, operator, or Agent instance.

Table 1: Related work by objective, evaluation scope, observed data, and intervention. Scope and labels are study-specific.

Study	Target	Scope	Observed data	Intervention
System identity and workflow				
Trace [2]	Model family	7 families; 3 frameworks	Terminal commands	Passive; follow-on probe
ProcGrep [6]	Agent workflow	10 coding Agents	Coding actions	Passive
LIDAR [7]	Base model	36 models; 2 frameworks	Coding behavior	Active probes
Known By Their Actions [8]	Browser model	14 models; 4 environments	UI events; timing	Passive
Cyber-agent behavior				
CyBiasBench [9]	Attack preference	5 Agents; 3 targets	HTTP requests	Task conditions
HABITS [3]	Human/Agent	Shopping sessions; 5 benchmarks	Browser actions; timing	Simulated and real traces
Attack correlation and trace retrieval				
Cross-Agent [10]	Campaign links	Synthetic sessions; framework probes	Events; text; timing	Logs and probes
This study	Same-task, same-mode retrieval	3 sources; mixed and Agent-only	Commands; launch gaps	Passive

Table 1 separates behavioral identity, Human/Agent profiling, and trace association targets. B/S/T is the shared feature schema; association and A/H scoring are independently evaluated applications with independent outputs.

3 Method: an interpretable B/S/T fingerprint

3.1 Pairwise representation

Let x and y be two traces. The historical label $z_{task} = 1$ indicates a shared benchmark task. The additional label $z_{mode} = 1$ requires both a shared task and the same recorded execution mode. The base evidence $E(x, y)$ contains command TF–IDF cosine and action-bigram TF–IDF cosine. Each trace also receives a windowed fingerprint: B summarizes local action organization and repetition; S summarizes novelty, revisits, and fragment reuse; T summarizes measured launch gaps when the source records the required time fields. Missing observations remain missing. No time value is inferred from event order or a workflow label.

The following view expands the primary pair path from event windows to the evidence passed to the association model. The auxiliary H/A readout remains separate.

The blocks address complementary properties of an observed trace. B1–B3 describe command or category diversity, entropy, and concentration, while B4–B8 describe action switching, conditional entropy, runs, and repetition. S1–S3 summarize novelty and revisits relative to the observed prefix; S4–S7 describe returns and fragment reuse, and S8 compares consecutive windows. Category transitions add a coarser view of action order. For example, the sequences AABABA and ABAABA have identical unigram and bigram counts, but their S4 return shares are 1 and 2/3. T describes the distribution and dependence of measured launch gaps, with per-category median gaps where available. The resulting fine-grained slots retain readable meanings while allowing the same per-trace schema to support different downstream tasks.

Figure 4 defines 22 window-level measurements: eight B, eight S, and six observed T measures. The fixed-length readout summarizes each feature column by its mean, standard deviation, and normalized-position trend. CTH and NYU use B/S because the evaluated trace views lack comparable event times; AIT supports measured process-launch gaps. For the independent A/H readout, each executable is first mapped to one of eight fixed operation categories, and B/S/T are then computed over that category sequence. This removes executable-name identity while preserving

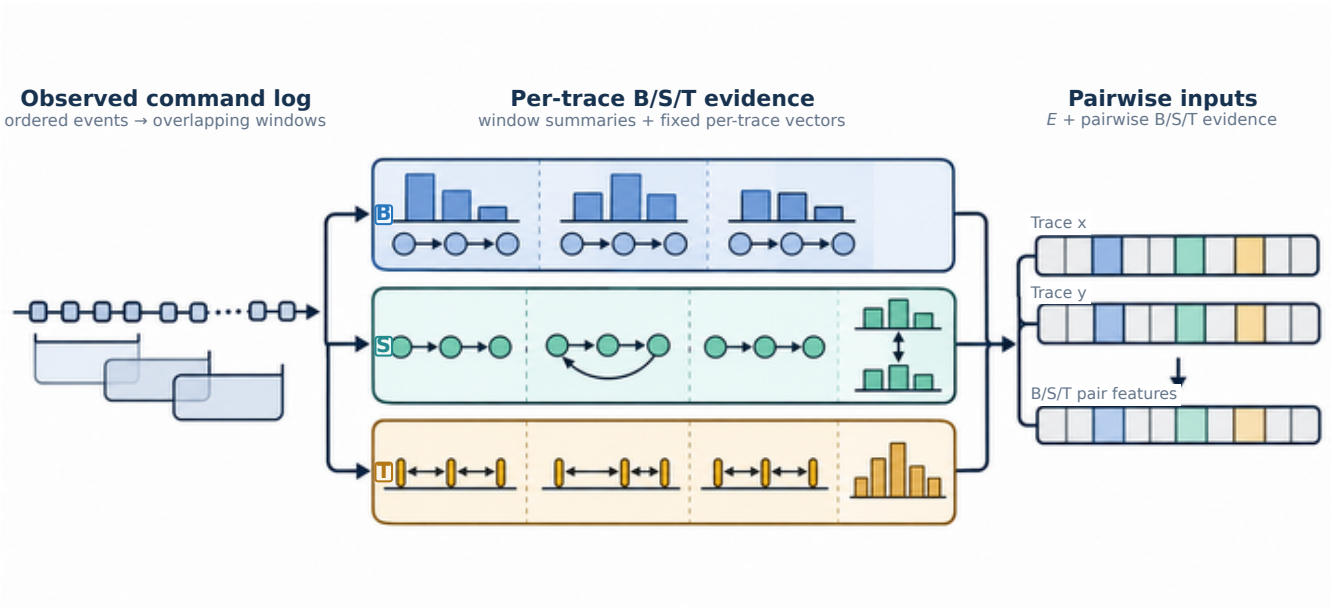


Figure 3: One tested construction of pairwise behavioral evidence. Ordered command events are divided into overlapping windows. B summarizes local organization, S summarizes action order and revisits, and T summarizes observed launch gaps. The association variant also evaluates tool-presence overlap and command/action-bigram evidence E. The separate A/H application instead uses category-normalized B/S/T and does not feed its score into pairwise association. T is omitted when launch timestamps are unavailable.

interpretable local organization, sequence, and launch-gap measurements. Raw-command B/S/T, command TF–IDF, and a fixed tool-presence vector are retained as reference controls. The A/H readout produces one continuous Agent-direction score per log; no such score is passed into trace association.

The matched CTH and NYU evaluations use B/S because their normalized trace views contain no event times. The upstream CTH archive retains Human terminal I/O timing and timestamped Agent events, but these records use different event units; this study has not established a validated, one-to-one mapping to a shared per-command start/end definition [12, 15]. We therefore do not treat the raw CTH archive as a comparable timing validation set. AIT records process-launch timestamps for all 109 included runs, supporting T1–T6; it does not provide matched event durations, so duration-dependent statistics are unavailable there. At eight-event windows, 84 of 85 CTH traces and all 109 AIT and 585 NYU traces have at least one finite value in each required B/S block. Coverage varies by measurement: in CTH, S2/S3 are finite in 42 traces and S8 in 26; in AIT, S4 is finite in 82, S5 in 107, and S8 in 101 traces. Sparse feature values are imputed using training-fold medians. The historical AIT implementation retained two unobserved duration columns as constant zeros after imputation; they could not contribute to its fitted scores. An E-only fallback is used only when a required block has no finite measurement for a trace.

For each available block, the pair representation appends the absolute difference between training-standardized trace summaries. In the selected conditional fusion, these differences are multiplied by command similarity before concatenation with E. B/S/T are evidence features for the pair classifier; the task, execution role, source, and path are not model inputs. The classifiers are RBF-SVM, a decision tree, and XGBoost. Their scores are not calibrated probabilities. If a required block has no finite measurement for either trace, the pair uses the same-fold E-only score and threshold; these fallback decisions remain in the evaluation.

The new same-mode evaluation uses direct fusion with the expanded per-trace B/S/T representation. It concatenates E with absolute differences of training-standardized continuous summaries and category vectors; B additionally contains the Jaccard overlap of fixed tool-presence sets. Eight-event windows admit a minimum of two events on CTH and four on AIT, with at most 128 events per trace. An empty tool union receives zero overlap. Incomplete required blocks fall back to E; when ranking a gallery containing a fallback pair, the entire gallery uses E scores. Length and tool overlap are also evaluated as a baseline control, and a B-internal ablation removes the tool-overlap feature.

3.2 Independent single-log Human/Agent score

The A/H application classifies one trace at a time using the category-normalized B/S/T representation. A balanced logistic model is fit inside each task-held-out or source-group-held-out fold, with missing-value imputation and feature scaling learned on training runs only. Its output is an uncalibrated, continuous Agent-direction score; the fixed 0.5 cutoff is used only to report BA, Human false-positive rate, and Agent recall. For explanation, each standardized feature's coefficient-weighted value is grouped by B, S, and T. These terms reconstruct that fold's logit exactly, but they are descriptive model contributions, not causal effects. This A/H result is an independent fingerprint application and is never used by the association model.



Figure 4: The 22-measurement B/S/T window schema. Glyphs illustrate definitions. Lowercase symbols denote commands and uppercase symbols coarse actions. n counts window events, m valid adjacent action pairs, U unique values, $\mathcal{H}$ prefix history, and Δ_t revisit lag. For observed launch-gap timing, g is a nonnegative gap in seconds and $\ell = \log(1+g)$. The A/H application maps command symbols to fixed operation categories before computing these measures; command-presence methods remain separate reference baselines.

4 Evaluation

4.1 Datasets and observed fields

Table 2 lists the eligible cohorts, recorded Human/Agent counts, and the log records used from each source. Each relation label identifies a benchmark task within its source.

Table 2: Evaluation cohorts. Green robots and blue people mark Agent and Human record counts; absent classes are omitted.

Source	Records / tasks	A / H counts	Dataset records
CTH mixture core	85 traces / 17 CTF tasks	51 + 34	Human terminal logs and Agent command-event logs
AIT audit logs [4]	109 runs / 12 tasks	48 + 61	Host audit records of process launches
NYU D-CIPHER [22]	585 runs / 160 challenges	585	Agent submission command histories

The CTH timing entries above describe the normalized trace views used by this study, not every field in the upstream archive. The public archive also contains Human terminal I/O timing and timestamped Agent events, but their event units differ; we have not validated a common per-command start/end mapping and therefore do not use them as a matched timing cohort [12, 15].

4.2 Pairing and validation

Historical same-task protocols. The CTH mixture analysis uses 17 task groups and 170 coupled gallery allocations at five Agent shares. Each query has one same-task and three different-task candidates, paired across shares. The share describes constructed galleries, not a resampled population proportion. The earlier matched CTH comparison contains 54,400 pair decisions; AIT contains 109 runs over 12 tasks and 43,600 pair decisions. Their label marks any shared task as positive, including cross-mode pairs. Detailed learner and feature results are reported in Appendix A.

Primary same-task, same-mode protocol. A positive pair shares both benchmark task and recorded mode. Each matched gallery contains one same-task, same-mode candidate, one same-task cross-mode candidate, and two different-task candidates, one from each mode. A no-match gallery replaces the positive with another different-task candidate. The same candidate identities support both the historical and primary labels. We use three task-held-out folds, 100 paired gallery draws, and depth-3 XGBoost with direct B/S/T fusion; CTH uses B/S and AIT uses B/S/T. Every arm uses identical outer galleries. Inner task folds select B/S/T blocks and calibrate thresholds. The guarded threshold arm additionally constrains inner pair FMR to 10% and inner task-macro no-match acceptance to the E baseline, then minimizes inner FSR. The grouped regularized arm selects its regularization strength using inner task-macro Hit@1. These retrospective cohorts were previously inspected, so all results are exploratory.

Metrics and uncertainty. We report task-macro Hit@1, balanced accuracy (BA), false-merge rate (FMR), false-split rate (FSR), and no-match acceptance: the fraction of no-match galleries with at least one accepted candidate. BA is $1 - (\mathrm{FMR} + \mathrm{FSR})/2$. Hit@1 is threshold-free; threshold changes affect link decisions but do not alter ranking. Detailed task-cluster intervals and calibration values are in the appendix. Repeated galleries and shared candidates are dependent, and these small cohorts do not support population-level claims.

5 Results

5.1 Same-task, same-mode retrieval

Table 3 reports the fixed B/S/T ablation on identical galleries. The pattern is not additive: on CTH, B alone raises Hit@1 by 5.45 points but also raises FMR by 2.32; S alone raises BA by 3.98 while raising FMR by 6.22. Full B/S improves Hit@1 by 3.33, alongside a 5.46-point rise in no-match acceptance. On AIT, B+T gives the highest Hit@1 (57.92%, +8.42), but FMR and no-match acceptance also rise by 3.72 and 13.11 points. Full B/S/T reaches 55.54% Hit@1 (+6.04) with a 7.75-point rise in no-match acceptance. These task-held-out results support a ranking/decision trade-off, not a uniform benefit from adding dimensions.

Table 3: Fixed B/S/T ablation for same-task, same-mode retrieval with depth-3 XGBoost. Rates are task-macro percentages; higher Hit@1/BA and lower FMR/FSR/no-match are favorable. Bold marks the best value within each dataset; colors compare with E, with changes below one point unshaded. T uses observed launch gaps. No-match is the share of no-match galleries with any accepted candidate.

Data	Features	Hit@1 ↑	BA ↑	FMR ↓	FSR ↓	No-match ↓
CTH	E	44.40	52.04	**6.47**	89.46	**18.45**
	E+B	**49.85**	53.79	8.79	83.64	23.81
	E+S	44.96	**56.02**	12.69	**75.26**	33.52
	E+B+S	47.74	54.87	7.78	82.48	23.91
AIT	E	49.50	57.05	**5.69**	80.22	**21.90**
	E+B	54.90	**63.84**	10.09	**62.23**	34.91
	E+S	46.16	58.06	6.95	76.94	26.49
	E+T	49.60	59.60	6.55	74.26	24.62
	E+B+S	53.72	61.21	8.74	68.84	32.99
	E+B+T	**57.92**	63.82	9.41	62.96	35.01
	E+S+T	49.21	58.44	7.70	75.42	28.09
	E+B+S+T	55.54	61.33	7.93	69.40	29.65

Inner-fold block selection improves Hit@1 but also raises false merges and no-match acceptance. The guarded threshold lowers these rates relative to unguarded selection, at the cost of more missed links; held-out no-match acceptance remains above E on both datasets. Grouped regularization favors a different operating point, particularly on AIT. Appendix Table 8 gives the full rates, paired intervals, and calibration details.

On the same outer candidates, a gallery-level cutoff moves AIT E+B+T from 9.41% to 6.88% FMR and from 35.01% to 27.86% no-match acceptance, while FSR rises from 62.96% to 66.47%; Hit@1 is unchanged. The CTH shift is smaller. Alternative S summaries improve ranking point estimates, but no tested S arm brings both false-acceptance measures below E on both datasets, so we retain the fixed ablation as the main result (Appendix Table 9).

5.2 Historical same-task mixture

Figure 2 shows the older same-task-only evaluation. From 0% to 100% Agent roles, action-bigram BA falls from 57.40% to 46.48%; at the fixed threshold, FSR rises as FMR falls. Hit@1 changes little. Because that label treats same-task cross-mode traces as positive, this figure describes a different retrieval relation and supplies motivation rather than evidence for the primary B/S/T result. Under the same historical label, CTH block effects vary by learner and show no common B/S improvement; full results and the post-hoc argument-enhanced analysis are in Appendix A.

5.3 Independent applications of the fingerprint

Agent-only cross-task retrieval. On the 111-challenge NYU snapshot protocol, each query is ranked against four traces from another challenge, with the matching D-CIPHER submission configuration as the positive. Adding B/S to E plus trace length raises full-trace gallery AUC by 2.19 points and Hit@1 by 2.93. On the shared 100-challenge, eight-command subset, AUC rises 1.19 points; on the 16-command subset, the AUC change is 0.26. This is exploratory evidence for cross-task configuration consistency within one Agent framework, not actor attribution. The separate frozen 160-challenge association protocol shows learner-dependent error trade-offs (Appendix B).

Single-log Human/Agent score. Table 4 reports the independent A/H application. Mapping command names to eight operation categories raises task-held-out BA from 68.8% to 75.8% relative to command-level B/S/T, although the task-clustered paired interval for this difference includes zero. Under source-group holdout, category-normalized B/S/T reaches 59.6% task-macro BA (95% CI [51.3, 67.0]); the command-level readout reaches 52.6%. Command TF–IDF and fixed tool presence are stronger references at 85.4% and 92.0%, respectively, so the current data do not establish source-independent A/H classification from the semantic fingerprint. The references use executable identity and are not the proposed category-normalized representation.

Figure 5 makes the use of an interpretable score concrete. The pair was selected by task and event-count difference before examining model outputs and is held out by task. Both logs contain 18 events. The recorded Human log

Table 4: Independent AIT Human/Agent readout on the same 109 recorded runs (61 Human, 48 Agent). Task and source BA are macro-averaged over the 12 troubleshooting tasks; intervals bootstrap tasks. AUROC, Human false-positive rate, and Agent recall pool held-out runs. Continuous scores are uncalibrated; the fixed 0.5 cutoff is used only for binary metrics.

Readout	Task BA [95% CI]	Source BA [95% CI]	AUROC	H false positive	A recall
B/S/T, command-level	68.8 [61.9, 75.7]	52.6 [42.9, 62.2]	0.54	26.2%	33.3%
B/S/T, category-normalized	**75.8 [69.7, 81.3]**	**59.6 [51.3, 67.0]**	**0.61**	26.2%	45.8%
Raw-command B/S/T + category summaries	89.4 [82.7, 95.3]	64.4 [55.6, 74.0]	0.72	16.4%	45.8%
Command TF–IDF reference	95.1 [92.4, 97.9]	85.4 [77.4, 92.7]	0.94	14.8%	85.4%
Fixed tool-presence reference	95.1 [91.7, 98.3]	92.0 [86.8, 96.9]	0.96	14.8%	97.9%

receives an Agent-direction score of 0.97, while the recorded Agent log receives 0.79. For the Human false positive, the grouped timing contribution is +3.91 logit units and the largest individual term is the share of launch gaps of at least ten seconds (+3.37). An analyst can see that timing drives the flag and inspect whether it reflects execution behavior or the capture workflow. This illustrates a review cue, not reliable attribution; the score is uncalibrated and cannot replace explicit provenance.

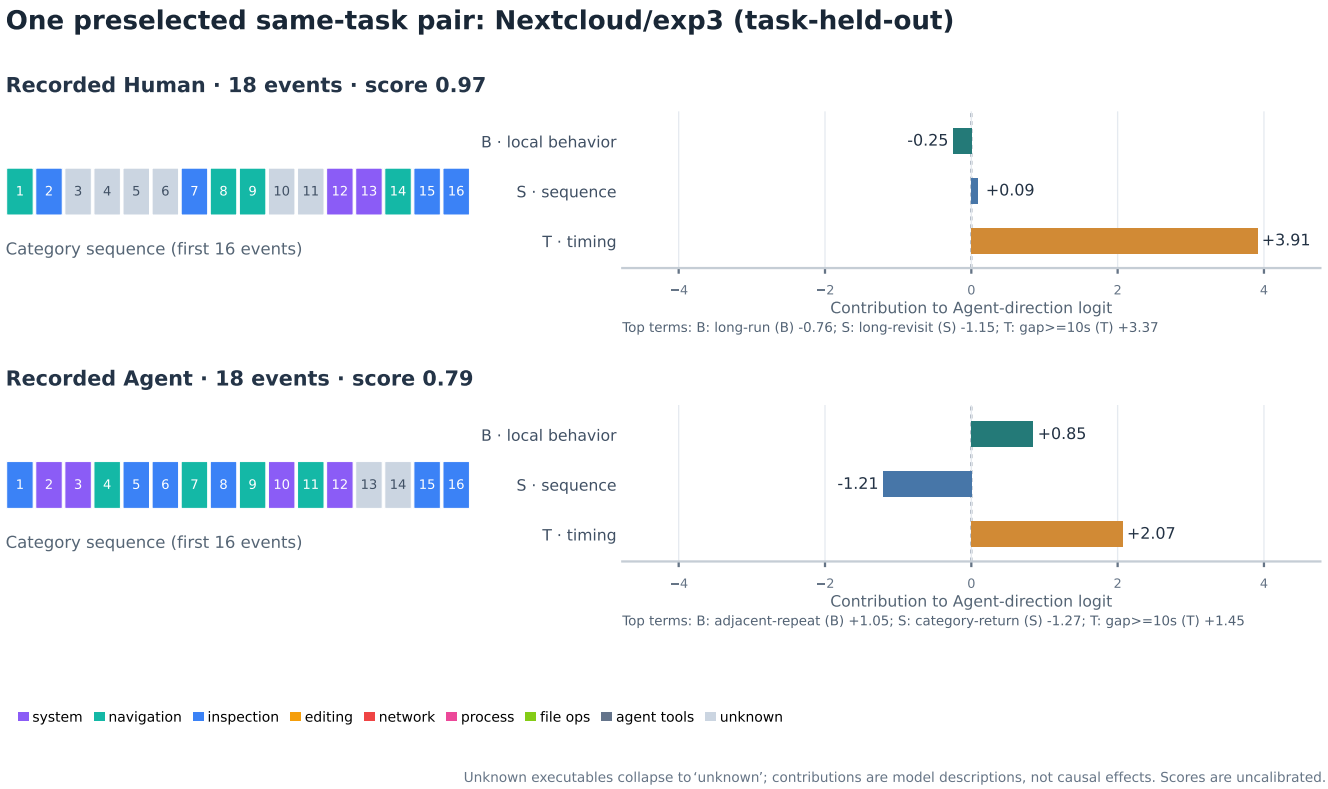


Figure 5: A preselected, same-task, event-count-matched pair in the task-held-out A/H readout. Left: the first 16 events mapped to operation categories. Right: signed B/S/T contributions to the Agent-direction logit; the fold-specific intercept plus these block totals reconstructs the score. The recorded Human is a false positive. The large timing contribution exposes a concrete reason to inspect the log. This score is independent of trace association.

6 Discussion

The fingerprint's generality resides in its measurement vocabulary: the same interpretable B/S/T definitions describe traces before a downstream analysis evaluates pairwise relatedness or assigns an Agent-direction score. Here, association compares per-trace summaries, while A/H scoring uses category-normalized features for one log. The matched Human false positive shows how a reviewer can trace a score to measured launch gaps and check the collection workflow. This illustrates a review affordance; the predictive limits of the A/H score are reported separately.

The A/H metrics also limit any transfer claim. Category-normalized B/S/T reaches task-held-out BA 75.8% but source-held-out BA 59.6%; the paired source-held-out change over command-level B/S/T is +6.9 points with a task-clustered interval that includes zero. The command-identity references do better on these AIT labels, which co-vary with task and collection workflow. Existing perturbation controls also do not establish that within-window timing order is the transferable cue: source-held-out BA is 52.6% for B/S/T and 49.9% after shuffling gaps within windows. The evidence supports a reusable, interpretable feature schema, while broad cross-source A/H accuracy remains unshown.

For trace association, B/S on CTH and B/S/T on AIT improve task-held-out Hit@1 point estimates, but false merges and accepted no-match galleries rise. Inner-fold block selection increases Hit@1 further, and the guarded cutoff reduces false acceptance at the cost of more missed links. No A/H score is used by these models. Thus the two applications provide separate evidence about the same representation; neither establishes an A/H-to-association gain.

The CTH composition curve addresses a related but different issue: under an older same-task-only label, a fixed threshold shifts from false merges toward missed links as the constructed gallery's Agent share changes. It does not establish that BST resolves the new relation. The detailed same-task ablation finds learner-dependent block effects, and the measured T increment is restricted to AIT launch gaps. In both sources, collection interface, task, and role can covary. The observed pair signal may therefore reflect behavior, capture process, trace volume, or their combination.

An exact-sequence audit also finds cross-task overlap in CTH: 30 of 85 held-out records match a training trace's command sequence, but every such sequence is only two to five events long; AIT has no exact cross-task command-sequence overlap. The CTH estimate is therefore sensitive to repeated short traces and should not be read as clean generalization to unseen behavior.

The NYU study asks whether one Agent framework's submission configurations remain retrievable across tasks. The A/H score uses an AIT troubleshooting label that co-varies with sources and workflows; source-held-out performance remains limited. All cohorts have been examined during development, and the evaluated data are not attack-response logs. Together, the experiments offer an exploratory demonstration of structural reuse across analysis objectives. They do not establish verified APT provenance, actor attribution, or universal transfer.

7 Conclusion

We present a fine-grained B/S/T fingerprint as a shared measurement layer for terminal-trace analysis and demonstrate two independent uses: pairwise trace association and single-log Human/Agent scoring. The A/H case shows how feature-level explanations can help analysts inspect a score, while the modest source-held-out result bounds claims about executor recognition. Association results expose ranking and false-acceptance trade-offs; A/H scores do not feed that task. Across the current CTF, troubleshooting, and Agent-submission cohorts, the results provide exploratory evidence that one interpretable feature vocabulary can support distinct analyses. They do not establish shared attackers, actor identity, or universal cross-source accuracy.

References

[1] F. Balassone et al. "Synthetic APTs: the Collapse of TTP-Based Attribution." arXiv:2606.07158, 2026. https://arxiv.org/abs/2606.07158

[2] Murali Ediga and Sudipta Chattopadhyay. "Trace: Unmasking AI Attack Agents Through Terminal Behavior Fingerprinting." arXiv:2605.01186, 2026. https://arxiv.org/abs/2605.01186

[3] Arka Dutta, Virinchi Roy Surabhi, Jiayue Tong, Dihan Dai, Olcay Boz, and Amila Weerasinghe. "Are LLM Shopping Agents Behaviorally Distinguishable from Human Shoppers? A Session-Level Study." In *Proceedings of the Second Workshop for REsearch on Agent Language Models (REALM 2026)*, pages 930–946. Association for Computational Linguistics, 2026. doi:10.18653/2026.realm-1.64. https://aclanthology.org/2026.realm-1.64/

[4] AIT AECID. "User behavior simulation for shell-based access using LLMs." GitHub repository, 2026. https://github.com/ait-aecid/llm-user-shell-sim

[5] Harold Booth, Bill Fisher, Ryan Galluzzo, and Tom Clancy. "Design Tenets for Agent Identity Management Systems." NIST Cybersecurity Insights, 2026. https://www.nist.gov/blogs/cybersecurity-insights/design-tenets-agent-identity-management-systems

[6] Hamidah Oderinwale. "Agent Trajectories as Programs: Fingerprinting and Programming Coding-Agent Behavior." arXiv:2606.16988, 2026. https://arxiv.org/abs/2606.16988

[7] Chuyi Wang et al. "Who Is Behind the Harness? Fingerprinting LLMs through Agentic Behavior." arXiv:2609.28559, 2026. https://arxiv.org/abs/2609.28559

[8] William Lugoloobi et al. "Known By Their Actions: Fingerprinting LLM Browser Agents via UI Traces." arXiv:2605.14786, 2026. https://arxiv.org/abs/2605.14786

[9] Taein Lim et al. "CyBiasBench: Benchmarking Bias in LLM Agents for Cyber-Attack Scenarios." arXiv:2605.07830, 2026. https://arxiv.org/abs/2605.07830

[10] SangJin Park et al. "Cross-Agent Campaign Attribution: Linking Asynchronous Attacks Across LLM Agents." arXiv:2607.18826, 2026. https://arxiv.org/abs/2607.18826

[11] "Honey for the Agent: Cyber Deception and Behavioral Fingerprinting." Zenodo record 20818246, 2026. https://zenodo.org/records/20818246

[12] Lyptus Research. "Cyber Task Horizons Data." 2026. https://github.com/lyptus-research/cyber-task-horizons-data

[13] M. M. McMahon and H. A. Sholl. "Influence of State Size on User Models in a Network Security System." 13th International Conference on Parallel and Distributed Computing Systems, 2000. https://www.researchgate.net/publication/282647993_Influence_of_State_Size_on_User_Models_in_a_Network_Security_System

[14] V. N. P. Dao, R. Vemuri, and S. J. Templeton. "Profiling Users in the UNIX OS Environment." Lawrence Livermore National Laboratory preprint UCRL-JC-141022, 2000. https://www.osti.gov/servlets/purl/15004770

[15] UK AI Safety Institute. "Inspect AI: Event Reference." https://inspect.aisi.org.uk/reference/inspect_ai.event.html

[16] A. Alsaheel et al. "ATLAS: A Sequence-Based Learning Approach for Attack Investigation." USENIX Security Symposium, 2021. https://www.usenix.org/conference/usenixsecurity21/presentation/alsaheel

[17] M. Milajerdi et al. "HOLMES: Real-Time APT Detection through Correlation of Suspicious Information Flows." IEEE Symposium on Security and Privacy, 2019. https://arxiv.org/abs/1810.01594

[18] Sutanu Kumar Ghosh et al. "OSTINATO: Cross-Host Attack Correlation through Attack Activity Similarity Detection." arXiv:2312.09321, 2023. https://arxiv.org/abs/2312.09321

[19] Sadegh M. Milajerdi et al. "POIROT: Aligning Attack Behavior with Kernel Audit Records for Cyber Threat Hunting." CCS, 2019. https://arxiv.org/abs/1910.00056

[20] Yi-Ting Huang et al. "A Cascade Approach for APT Campaign Attribution in System Event Logs: Technique Hunting and Subgraph Matching." arXiv:2410.22602, 2024. https://arxiv.org/abs/2410.22602

[21] H. Yang et al. "InterCode: Standardizing and Benchmarking Interactive Coding with Execution Feedback." NeurIPS, 2023. https://arxiv.org/abs/2306.14898

[22] NYU-LLM-CTF. "Leaderboard Submissions." GitHub repository, revision recorded in the frozen evaluation manifest. https://github.com/NYU-LLM-CTF/leaderboard_submissions

A Supplementary CTH and external results

A.1 Historical same-task ablations and ranking

The following tables use the historical label that any same-task pair is positive. They provide context for Figure 2 and are separate from the primary same-task, same-mode evaluation.

Table 5: Full feature-block ablation for same-challenge association.

Learner	Feature arm	BA (%)	FMR (%)	FSR (%)
CTH mixed roles (17 tasks; 54,400 pair decisions per arm)				
RBF-SVM	**E (baseline)**	**52.57**	**6.52**	**88.35**
	E+B	54.08 (+1.51)	14.00 (+7.49)	77.83 (-10.51)
	E+S	52.45 (-0.12)	13.57 (+7.05)	81.54 (-6.81)
	E+B+S	53.95 (+1.38)	13.50 (+6.99)	78.60 (-9.74)
Decision tree	**E (baseline)**	**52.80**	**6.59**	**87.81**
	E+B	50.31 (-2.49)	0.93 (-5.66)	98.44 (+10.63)
	E+S	51.79 (-1.01)	0.89 (-5.70)	95.53 (+7.72)
	E+B+S	50.00 (-2.80)	0.00 (-6.59)	100.00 (+12.19)
XGBoost	**E (baseline)**	**56.06**	**5.97**	**81.91**
	E+B	52.75 (-3.31)	13.13 (+7.16)	81.37 (-0.54)
	E+S	56.10 (+0.04)	14.47 (+8.50)	73.34 (-8.57)
	E+B+S	53.68 (-2.38)	15.43 (+9.46)	77.21 (-4.70)
AIT audit logs (12 tasks; XGBoost; 43,600 pair decisions per arm)				
XGBoost	**E (baseline)**	**50.12**	**0.00**	**99.76**
	E+B	50.94 (+0.82)	2.78 (+2.78)	95.34 (-4.42)
	E+S	53.18 (+3.06)	3.71 (+3.71)	89.94 (-9.82)
	E+T	52.42 (+2.30)	4.84 (+4.84)	90.32 (-9.44)
	E+B+S	50.94 (+0.82)	5.28 (+5.28)	92.84 (-6.92)
	E+B+T	51.37 (+1.25)	5.24 (+5.24)	92.02 (-7.74)
	E+S+T	52.01 (+1.89)	3.53 (+3.53)	92.45 (-7.31)
	E+B+S+T	51.92 (+1.80)	5.37 (+5.37)	90.79 (-8.97)

Cells show task-macro percentages and changes from E in parentheses. Teal and red indicate favorable and unfavorable changes; unshaded changes are below one point. T summarizes observed launch gaps in AIT.

A.2 Same-task, same-mode feature ablation

Table 3 reports the core factorial ablation. The supplementary table below adds length/presence and B tool-overlap controls.

A.3 Block selection and threshold operating points

These exploratory comparisons use training tasks for block selection and threshold calibration; no outer-fold labels are used during selection.

A.4 S-dimension and gallery-level threshold follow-up

To separate feature effects from cutoff effects, we compare the original pairwise threshold with a gallery-level guard on the same outer candidate identities. The guard constrains inner task-macro pair FMR and no-match acceptance to the E reference, then chooses the least restrictive feasible cutoff. It improves the AIT false-acceptance operating point but increases missed links; outer no-match rates still exceed E. Scalar S summaries, revisit/reuse statistics, smoothed category transitions, and category n-grams raise ranking point estimates on some folds, but their error trade-offs vary by source. These results are exploratory, and the inner constraints did not reliably transfer to held-out tasks.

Table 6: Candidate-ranking performance for the predeclared feature comparisons.

Learner	Scored evidence	Hit@1 (%)	Gallery AUC (%)	Pair AP (%)
CTH mixed roles: E+B+S compared with E				
RBF-SVM	**E (baseline)**	**33.75**	**58.73**	**39.78**
	E+B+S	38.45 (+4.70)	62.22 (+3.49)	42.66 (+2.88)
Decision tree	**E (baseline)**	**34.91**	**60.30**	**35.21**
	E+B+S	36.27 (+1.37)	59.77 (-0.53)	36.44 (+1.23)
XGBoost	**E (baseline)**	**34.40**	**58.92**	**40.97**
	E+B+S	37.01 (+2.61)	61.73 (+2.81)	41.06 (+0.08)
AIT XGBoost: T increment is compared with E+B+S				
XGBoost	**E (baseline)**	**40.20**	**63.34**	**38.25**
	E+B+S	33.33 (-6.88)	57.87 (-5.47)	32.60 (-5.65)
	E+B+S+T1–T6†	34.01 (+0.68)	58.80 (+0.93)	34.11 (+1.50)
NYU Agent-only external evaluation: frozen E+B+S compared with E				
RBF-SVM	**E (baseline)**	**39.73**	**65.75**	**42.77**
	E+B+S	38.67 (-1.06)	65.56 (-0.18)	41.35 (-1.41)
Decision tree	**E (baseline)**	**36.92**	**63.94**	**35.29**
	E+B+S	31.33 (-5.59)	56.80 (-7.13)	32.29 (-3.00)
XGBoost	**E (baseline)**	**34.74**	**61.29**	**39.55**
	E+B+S	37.94 (+3.20)	63.10 (+1.82)	41.62 (+2.07)

Cells give absolute task-macro percentages and signed paired changes in parentheses. CTH and NYU changes compare E+B+S with E; the AIT † row gives the incremental change over E+B+S. Higher ranking values are favorable. Pale-to-deeper teal/red backgrounds encode fixed absolute-change bands of 1–< 3, 3–< 8, and ≥ 8 points; changes below 1 point are unshaded. Colors do not encode statistical significance. If any pair in a four-candidate gallery lacks a required block, all four ranking scores use the matched E score. Complete factorial ranking estimates are in the appendix; task-cluster intervals are in the experiment record.

Table 7: Length/presence and tool-overlap controls for same-task, same-mode retrieval. Rates are task-macro percentages; colors compare with E. B^- omits tool-presence overlap.

Data	Features	Hit@1	BA	FMR	FSR	No-match
CTH	E	44.40	52.04	6.47	89.46	18.45
	E+length+presence	48.71	57.47	3.88	81.18	14.40
	E+length+presence+B+S	46.79	56.61	7.87	78.91	25.38
	$E+B^-+S$	47.43	52.80	10.79	83.61	26.39
AIT	E	49.50	57.05	5.69	80.22	21.90
	E+length+presence	50.10	61.22	7.57	69.98	27.21
	E+length+presence+B+S+T	55.66	62.95	8.68	65.42	31.64
	$E+B^-+S+T$	50.13	58.93	7.45	74.69	27.87

Table 8: Exploratory operating-point variants for same-task, same-mode retrieval. Task-macro rates are percentages. E and fixed full-profile reference results are in Table 3; block selection and thresholds use training tasks only. Teal and red compare each variant with E. No-match is the share of no-match galleries with any accepted candidate.

Data	Method	Hit@1 ↑	BA ↑	FMR ↓	FSR ↓	No-match ↓
CTH	Inner-selected blocks	48.60	56.00	11.53	76.47	33.49
	Guarded threshold	48.60	54.70	10.54	80.06	29.73
	Grouped regularized	45.04	54.01	8.35	83.64	23.13
AIT	Inner-selected blocks	57.92	63.82	9.41	62.96	35.01
	Guarded threshold	57.92	63.32	6.88	66.47	27.86
	Grouped regularized	54.58	64.52	12.15	58.81	40.79

Table 9: S-dimension and gallery-threshold follow-up on the same outer candidates. Base is E+B for CTH and E+B+T for AIT. Pair cutoff uses the original inner pairwise FMR rule; gallery guard uses inner task-macro FMR and no-match constraints. S variants append the named S features to Base. Rates are task-macro percentages; higher Hit@1/BA and lower FMR/FSR/no-match are favorable. Bold marks the best value per dataset and metric. The 2× hard-negative arms match their parent arms at reported precision and are omitted.

Data	Profile / cutoff	Hit@1 ↑	BA ↑	FMR ↓	FSR ↓	No-match ↓
CTH	E	44.40	52.04	**6.47**	89.46	**18.45**
	Base, pair cutoff	49.85	53.79	8.79	83.64	23.81
	Base, gallery guard	49.85	53.24	8.69	84.84	22.99
	Full S, pair cutoff	47.74	**54.87**	7.78	**82.48**	23.91
	Full S, gallery guard	47.74	54.35	7.58	83.72	22.98
	S scalar summaries	49.96	54.12	7.01	84.74	20.06
	S revisit / reuse	**50.87**	53.47	8.34	84.72	21.18
	S + smoothed transitions	50.42	54.12	7.01	84.74	20.06
	S + category n-grams	50.03	54.06	7.14	84.74	20.65
	Nested selection	47.60	53.10	8.92	84.88	24.34
AIT	E	49.50	57.05	**5.69**	80.22	**21.90**
	Base, pair cutoff	**57.92**	63.82	9.41	**62.96**	35.01
	Base, gallery guard	**57.92**	63.32	6.88	66.47	27.86
	Full S, pair cutoff	55.54	61.33	7.93	69.40	29.65
	Full S, gallery guard	55.54	61.29	6.32	71.10	25.82
	S scalar summaries	56.73	61.91	7.36	68.81	28.83
	S revisit / reuse	56.39	63.64	7.38	65.34	28.75
	S + smoothed transitions	57.22	61.24	7.50	70.02	28.69
	S + category n-grams	56.85	**64.13**	8.22	63.51	30.29
	Nested selection	56.93	61.63	7.45	69.28	28.82

An exact-overlap audit checks ordered command, mapped-action, and joint sequences after the 128-event cap, without semantic normalization. Under the frozen CTH folds, 30 of 85 test records (35.3%) have a command sequence also present in training under another task; these matches comprise six cross-task sequence groups and all are two to five events long. Mapped-action sequences overlap for 35 of 85 records across seven groups. AIT has no exact cross-task overlap for command, action, or joint sequences. This audit identifies repeated short observations, not near-duplicate behavior or causal leakage; it bounds how strongly task-held-out CTH results support unseen-behavior generalization.

A.5 Uncertainty for exploratory comparisons

The original CTH XGBoost argument-enhanced B/S contrast has a nominal interval $[+1.17, +12.24]$ BA points and a max-T interval $[-0.75, +13.24]$. In the same-mode evaluation, full-profile Hit@1 changes have paired task intervals $[-5.07, +11.13]$ on CTH and $[-0.22, +11.17]$ on AIT. AIT B+T has $[+4.50, +12.40]$ in the multi-arm comparison. The updated NYU full-trace AUC and Hit@1 changes have intervals $[+1.26, +3.16]$ and $[+1.53, +4.39]$. The eight-command subset yields $[+0.15, +2.31]$ and $[-0.04, +2.63]$, respectively; the 16-command AUC interval is $[-0.95, +1.48]$. All intervals are percentage points, are exploratory, and depend on the stated task-cluster resampling.

For inner-fold block selection versus E, the paired Hit@1 change is +4.19 points on CTH ($[-2.64, +10.81]$) and +8.42 on AIT ($[+4.50, +12.40]$). The guarded threshold preserves Hit@1 exactly. Relative to the unguarded selected arm, it changes FMR by −0.98 points on CTH ($[-1.76, -0.32]$) and −2.52 on AIT ($[-3.95, -1.21]$); no-match acceptance changes by −3.76 ($[-6.73, -1.32]$) and −7.15 ($[-10.67, -3.83]$), respectively. FSR rises by +3.59 points on CTH ($[0.00, +9.53]$) and +3.51 on AIT ($[+1.43, +5.74]$). One CTH calibration fold selects an infinite cutoff, rejecting every non-fallback pair. The inner-fold constraints therefore do not imply the same operating rates on held-out tasks.

Figure 8 displays paired changes by role type, while Table 12 gives exact XGBoost values by mixture share. The external table reports learner-specific absolute results and the paired error changes; its intervals cluster the 160 held-out challenge groups.

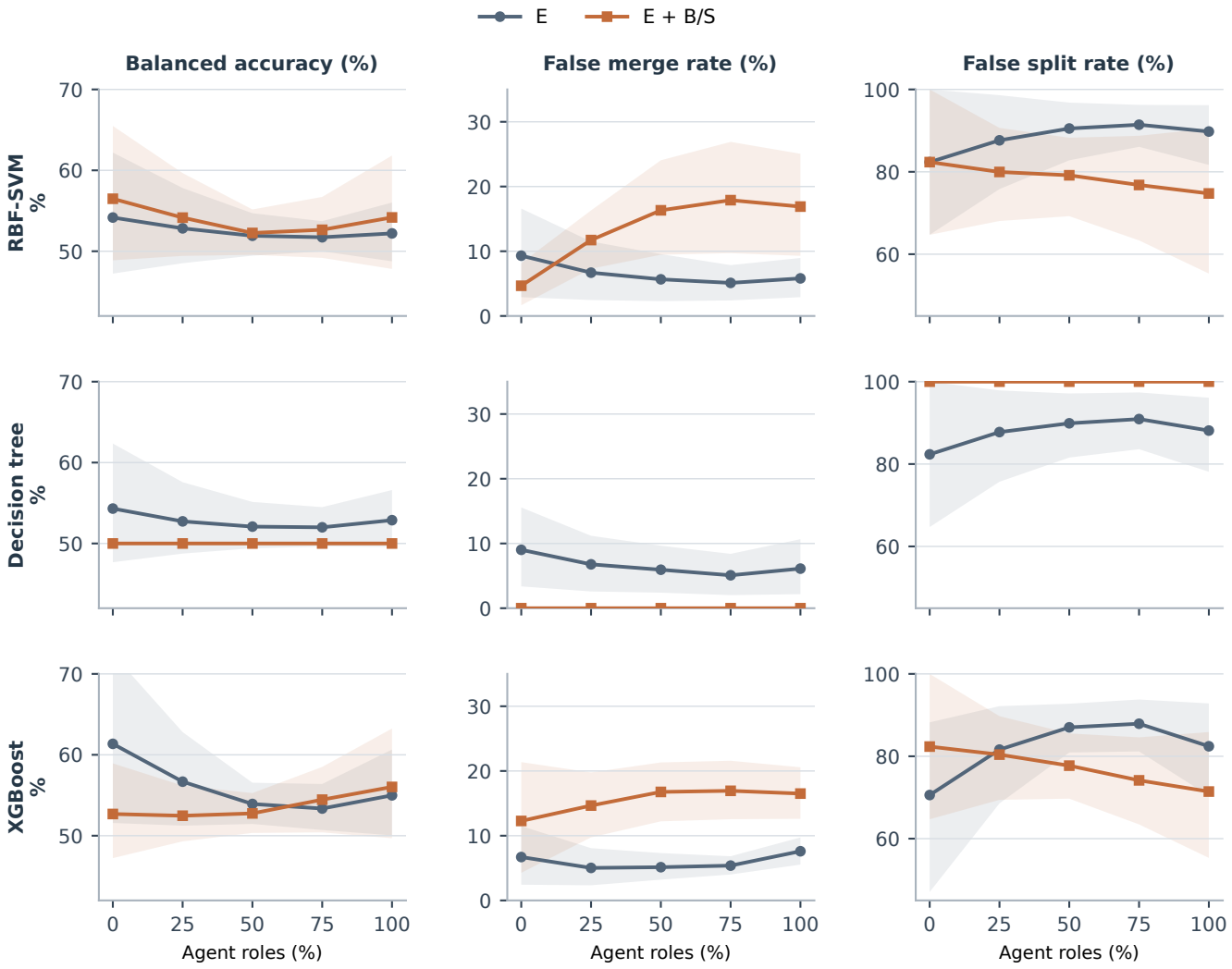


Figure 6: Matched CTH comparison across Agent-role shares. Lines show E and E+B/S; bands give task-cluster intervals.

A.6 Complete factorial ranking ablation

The main ranking table shows the predeclared contrasts; Table 13 preserves every CTH and AIT feature combination. For ranking, any gallery with at least one unavailable pairwise block is scored entirely with its matched E baseline so all four candidates share one scoring scale.

B Frozen external evaluation protocol

The NYU task manifest was frozen before fitting. Four public D-CIPHER snapshots from the official NYU-LLM-CTF leaderboard-submissions repository were included [22]. The exact source revision and hashes are in the frozen evaluation manifest. Challenge groups overlapping the CTH development pool and one duplicate command sequence were removed. The remaining cohort in Table 2 was partitioned by a fixed hash into three retrieval galleries. For each run query, 100 deterministic draws selected one other run from the same challenge and one run from each of three different challenges in the same partition. All three classifiers and both feature arms used the identical 234,000-pair list. Training used only the CTH development task groups and all available training pairs; model parameters and thresholds were not selected from NYU outcomes. The CTH-derived threshold did not preserve its target FMR on the external source.

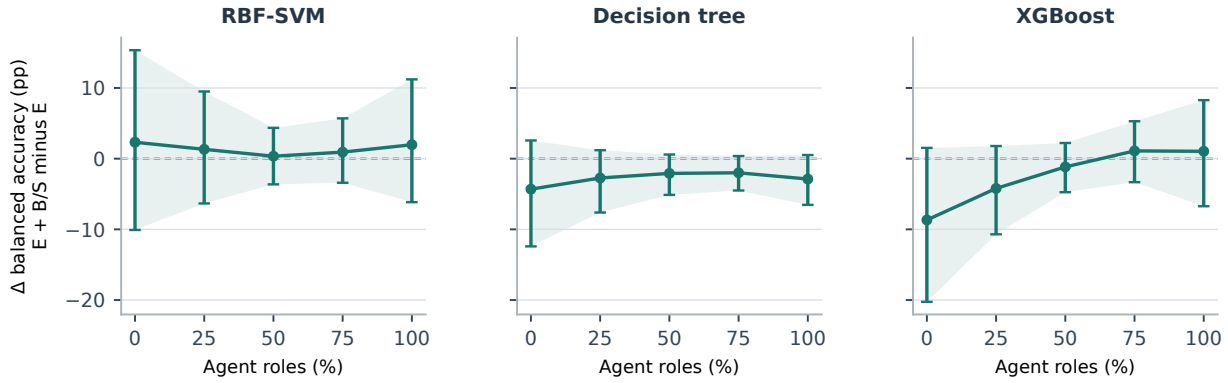


Figure 7: Paired BA change (E+B/S minus E) across CTH shares. Bars give task-cluster intervals.

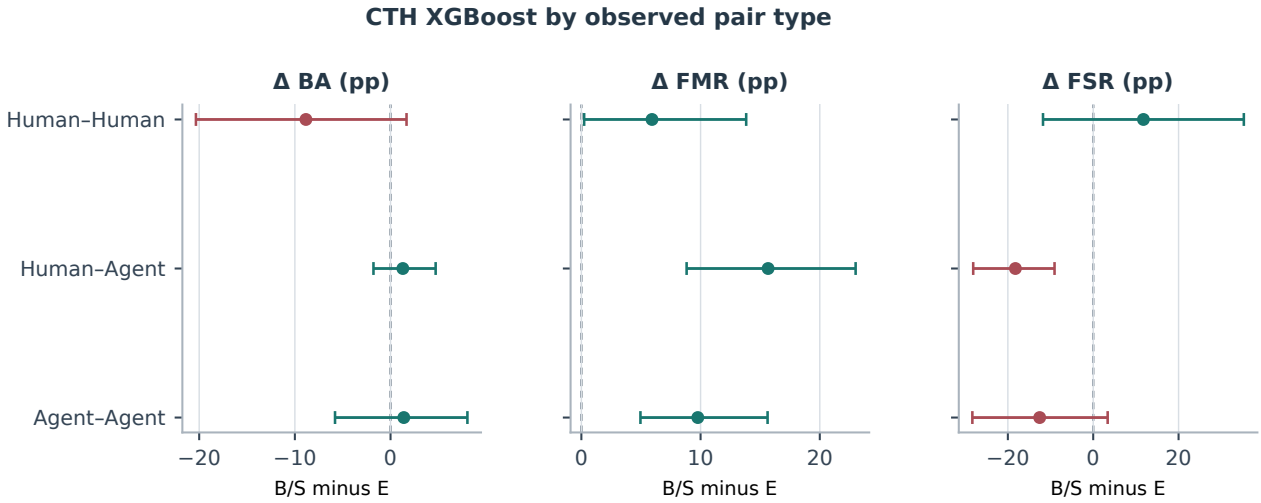


Figure 8: CTH XGBoost change (E+B/S minus E) by pair type and metric. Points are share-equal means over the eligible proportions; bars show paired task-cluster 95% intervals. HH, HA, and AA are evaluation roles, not attack identities.

Snapshot composition. The evaluation contains four submissions of the same D-CIPHER framework. Table 14 reports how the raw transcripts map to eligible traces and distinct challenge groups. Challenge counts overlap across snapshots; the total is deduplicated.

The official repository and snapshot metadata identify challenge filenames and run configuration. A shared challenge name is only the relation proxy. The four snapshots all belong to one Agent framework; this dataset cannot test Human–Agent associations or activity/actor attribution. Error transitions are counts of repeated pair decisions, not independent events: XGBoost corrected 11,241 earlier false merges and introduced 37,348 new false merges, while it corrected 19,708 earlier false splits and introduced 4,329 new false splits. These counts explain the rate changes but do not increase the independent sample size.

C Pair-type support and interpretation

The CTH type-specific estimates reuse the outer predictions. HH is supported at 0–75% Agent roles, HA at 25–75%, and AA at 25–100%. Each type is averaged equally across the shares where it exists and then across tasks. At 50%, the XGBoost gallery contributes 662 positive and 2,004 negative HH pair decisions, 1,396 positive and 4,161 negative HA decisions, and 662 positive and 1,995 negative AA decisions. Pair decisions share queries, candidate tasks, and task clusters; the counts are denominators, not independent sample sizes. The interval widths and overlapping task support preclude interpreting the three rows as independent replications.

Table 10: Frozen NYU D-CIPHER AA challenge-retrieval comparison. The challenge filename defines the same-task proxy. Changes are E+B/S minus E; intervals cluster by 160 task groups.

Classifier	BA: E	BA: E+B/S	ΔBA [95% CI]	FMR: E+B/S	ΔFMR	ΔFSR
RBF-SVM	58.16	49.93	−8.23 [−10.22, −6.28]	0.13	-26.45	+42.91
Decision tree	49.69	53.33	+3.64 [+1.97, +5.37]	8.54	+4.69	-11.98
XGBoost	51.16	56.87	+5.71 [+3.63, +7.73]	25.39	+14.67	-26.10

Table 11: CTH XGBoost error changes by pair type, averaged equally over the role shares where the type exists. HH, HA, and AA are trace-role pairs, not activity or actor labels.

Pair type	ΔBA [95% CI]	ΔFMR	ΔFSR
Human–Human	−8.84 [−20.34, +1.67]	+5.92	+11.76
Human–Agent	+1.29 [−1.78, +4.71]	+15.66	-18.23
Agent–Agent	+1.39 [−5.80, +8.03]	+9.77	-12.54

Table 12: CTH XGBoost results by exact Agent-role share. Values are task-macro means over the same paired candidates; ΔBA intervals preserve task clustering.

Agent roles	BA: E	BA: E+B/S	ΔBA [95% CI]	FMR: E+B/S	FSR: E+B/S
0%	61.35	52.68	−8.67 [−20.26, +1.53]	12.28	82.35
25%	56.68	52.47	−4.21 [−10.70, +1.79]	14.66	80.40
50%	53.92	52.76	−1.16 [−4.74, +2.22]	16.76	77.72
75%	53.36	54.45	+1.10 [−3.33, +5.29]	16.94	74.15
100%	54.99	56.03	+1.04 [−6.73, +8.28]	16.51	71.43

Table 13: Complete factorial ranking ablation. Values are absolute percentages followed by changes from E.

Source	Learner	Feature arm	Hit@1	Gallery AUC	Pair AP
CTH	**RBF-SVM**	**E (baseline)**	**33.75**	**58.73**	**39.78**
		E+B	36.93 (+3.18)	60.63 (+1.90)	40.90 (+1.13)
		E+S	35.52 (+1.77)	61.67 (+2.94)	40.57 (+0.79)
		E+B+S	38.45 (+4.70)	62.22 (+3.49)	42.66 (+2.88)
	Decision tree	**E (baseline)**	**34.91**	**60.30**	**35.21**
		E+B	27.49 (-7.41)	53.62 (-6.69)	30.90 (-4.31)
		E+S	32.08 (-2.83)	56.14 (-4.17)	34.07 (-1.15)
		E+B+S	36.27 (+1.37)	59.77 (-0.53)	36.44 (+1.23)
	XGBoost	**E (baseline)**	**34.40**	**58.92**	**40.97**
		E+B	33.53 (-0.87)	58.81 (-0.11)	38.98 (-1.99)
		E+S	36.95 (+2.55)	61.49 (+2.57)	40.68 (-0.30)
		E+B+S	37.01 (+2.61)	61.73 (+2.81)	41.06 (+0.08)
AIT	**XGBoost**	**E (baseline)**	**40.20**	**63.34**	**38.25**
		E+B	34.20 (-6.00)	57.47 (-5.86)	33.35 (-4.90)
		E+S	37.05 (-3.15)	60.31 (-3.02)	35.80 (-2.45)
		E+T1–T6	39.02 (-1.19)	62.78 (-0.56)	36.73 (-1.52)
		E+B+S	33.33 (-6.88)	57.87 (-5.47)	32.60 (-5.65)
		E+B+T1–T6	35.53 (-4.67)	59.46 (-3.88)	33.96 (-4.29)
		E+S+T1–T6	38.02 (-2.18)	61.70 (-1.64)	36.76 (-1.49)
		E+B+S+T1–T6	34.01 (-6.19)	58.80 (-4.53)	34.11 (-4.14)

Each change is a paired task-level percentage-point difference from the same learner/source E baseline; the AIT +T increment is separately shown in the main ranking table. Teal/red shading uses the fixed magnitude bands described in Table 6; colors do not encode significance. CTH includes 17 task clusters and 54,400 pair decisions per arm; AIT includes 12 clusters and 43,600 decisions per arm. Gallery ranking uses whole-gallery E fallback when any candidate lacks a required block. Intervals, tie counts, and fallback details are recorded in the accompanying experiment Markdown and JSON.

Table 14: NYU D-CIPHER snapshot composition. Challenge counts overlap across snapshots; aggregate eligible counts are reported in Table 2.

Model snapshot	Raw transcripts	Eligible traces	Eligible challenges
Claude 3.5 Sonnet	194	147	147
Claude 3.7 Sonnet	200	128	128
GPT-4.1	196	153	153
GPT-4o	197	157	157

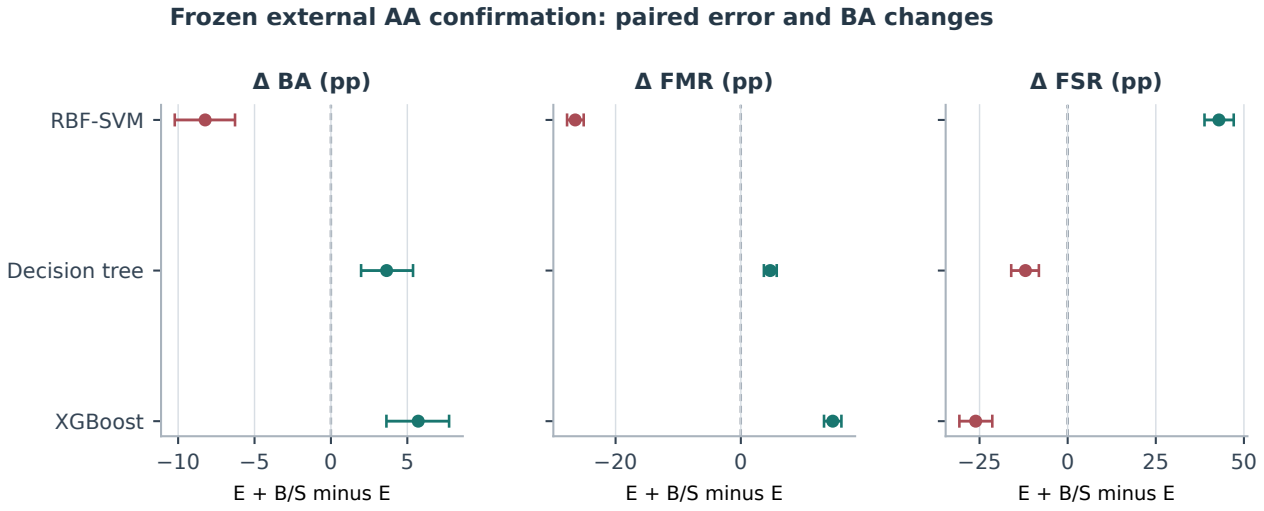


Figure 9: Frozen NYU AA evaluation: paired changes in BA, FMR, and FSR for E+B/S relative to E. The vertical dashed reference is zero; intervals cluster 160 challenge groups.

D Fingerprint definitions and supplementary evaluations

Table 15 lists the feature definitions in Figure 4. The supplementary evaluations examine feature reduction, ranking loss, cross-challenge snapshot retrieval, and observed timing in AIT.

For a window of n commands, let $p(c)$ be the empirical command frequency, P the multiset of adjacent pairs with known action labels, and $N(a, b)$ and $N(a)$ the corresponding pair and outgoing-action counts. B1 is $|\{c\}|/n$, B2 is $-\sum_c p(c) \log_2 p(c)$, and B3 is $\max_c p(c)$. B4 is the fraction of pairs in P whose actions differ, B5 is $-\sum_{a,b} N(a, b)|P|^{-1} \log_2(N(a, b)/N(a))$, B6 is the longest known-action run divided by n, B7 is $1 - |\text{unique}(P)|/|P|$, and B8 is the fraction of adjacent equal command names. Pair-dependent quantities are missing when their denominators are zero.

For each command in S, the history contains previously observed commands and fragments. S1 is the fraction of first occurrences. Among repeated commands, S2 is the fraction whose preceding occurrence is at least two positions earlier, and S3 averages $\log(1 + \Delta)/\log(129)$ over their revisit lags Δ. S4 counts returns a, b, a among known-action triplets with $a \neq b$; S5 compares the first and fourth known actions in each valid four-event segment. S6 and S7 count command bigrams and trigrams already observed in the prefix, relative to their respective valid fragments. S8 is the Jensen–Shannon divergence between the known-action distributions of the current and preceding nonempty windows. A statistic is missing when its required comparisons or preceding window are unavailable.

For nonnegative adjacent launch gaps g, T1 and T2 are the median and interquartile range of $\log(1 + g)$. T3 is $(\sigma_g - \mu_g)/(\sigma_g + \mu_g)$, with zero when the denominator is zero. T4 and T5 are the fractions of gaps at most one second and at least ten seconds. T6 is the lag-one correlation of log gaps and requires at least three gaps and nonzero variance in both paired sequences. The separate category-time vector uses the median log gap between successive observed launches within each category. Insufficient gaps remain missing.

Table 15: Window-level feature definitions. A value is missing when its required observations are absent or insufficient.

Block	ID	Measurement
B	B1–B3	Distinct command share; Shannon entropy of command names; share of the dominant command.
B	B4–B6	Fraction of known adjacent action pairs that switch group; conditional entropy of the next action; longest single-action run divided by window length.
B	B7–B8	One minus the unique mapped adjacent-action-pair share; fraction of adjacent command names that are identical.
S	S1–S3	First-seen command share; fraction of repeated commands whose previous occurrence is at least two positions back; mean normalized log revisit span.
S	S4–S5	A-B-A action return share among valid triplets; share of valid four-action segments whose first and fourth action match.
S	S6–S8	Reused adjacent command-bigram share; reused command-trigram share; Jensen–Shannon divergence between current and preceding window action distributions.
T	T1–T3	Median log(1+gap); interquartile range of log gaps; burstiness $(\sigma - \mu)/(\sigma + \mu)$ over nonnegative raw gaps.
T	T4–T6	Fraction of gaps at most one second; fraction at least ten seconds; lag-one correlation of log gaps when at least three valid gaps exist.

D.1 Exploratory single-measure reduction

A nested diagnostic on the 17 CTH development tasks compared full B/S with 16 candidates, each dropping one window measurement. Inner task folds selected the deletion under the existing FMR-at-most-10% threshold rule. The selected removals differed across folds (B6, S7, and B2). Both ranking and decision metrics use the E score on pairs with a missing block. Against full B/S, the compact version changed BA by -0.59 pp (95% task interval $[-2.02, +0.71]$), Hit@1 by -0.24 pp $[-3.13, +2.15]$, gallery AUC by -0.32 pp $[-1.92, +1.03]$, and pair AP by -0.14 pp $[-1.75, +1.38]$. The results leave the value of deleting an individual measure uncertain.

D.2 CTH ranking loss

On the same 54,400 candidate pairs, we compared a logistic pair classifier with a logistic ranker trained on feature differences between positives and negatives for the same query. Both used identical training pairs and task weights. The inner folds selected regularization by gallery AUC and calibrated thresholds under the existing FMR rule. With E alone, changing the loss changed gallery AUC by -0.49 pp $[-1.11, +0.02]$. With E+B/S, it changed AUC by -5.90 pp

$[-12.27, -1.06]$ and Hit@1 by -6.97 pp $[-12.71, -2.17]$. The lowest held-out FMR among these four logistic arms was 13.83%, above the 10% validation target.

D.3 Cross-challenge snapshot retrieval

This separate evaluation uses 111 NYU challenges with one trace from each of four D-CIPHER submissions (444 traces). Each query is matched against four runs from one different challenge, one per snapshot. Only the matching snapshot is positive. We hold out entire challenges, train a fixed logistic ranker on the other challenges, and resample challenge IDs jointly in both roles for the intervals. E with trace length yields 67.66% gallery AUC and 44.27% Hit@1. Adding B/S raises AUC by 1.19 pp $[+0.44, +1.97]$ and Hit@1 by 1.52 pp $[+0.42, +2.68]$.

On 100 challenges with eight or more observed commands per run, the AUC increment is $+1.08$ pp $[+0.25, +1.96]$ with full traces and $+0.12$ pp $[-0.82, +1.11]$ with eight-command prefixes. A secondary check on 80 challenges gives -0.72 pp $[-1.78, +0.34]$ with 16-command prefixes. These contrasts concern known submission configurations in a shared framework. Snapshot labels include any differences in their configurations.

D.4 Observed AIT behavior and timing



Figure 10: Two AIT traces from the same troubleshooting task, visualized against the B/S/T schema over variable-length windows. The shared blue–amber scale shows standardized values for display. Hatched cells mark missing observations.

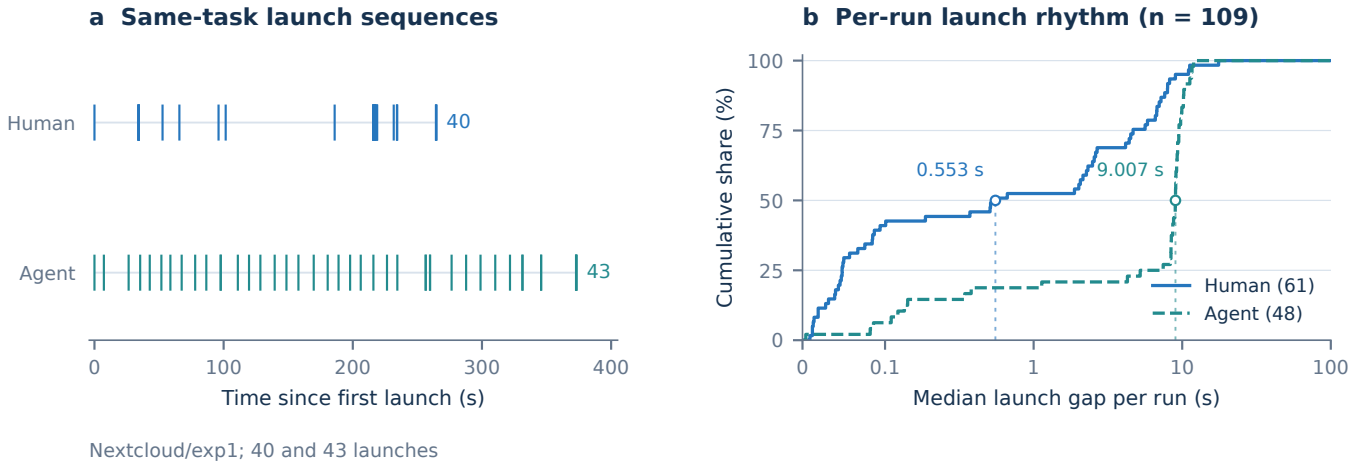


Figure 11: Observed launch-gap timing behind T1–T6. Left, real launch timestamps from the two traces in Figure 10, aligned to their own first observed launch with a shared seconds scale. Right, empirical distributions of one median launch gap per run over all 109 AIT runs. The scale is logarithmic above 0.1 seconds and preserves zero gaps. Each run contributes at most its first 128 login-associated events.

Sensitivity to within-window order. On the 109-run cohort, the existing source-held-out B/S score changes from 35.8% task-macro BA to 40.3% when eight-event windows are permuted; the paired task interval includes zero. For B/S/T, shuffling gaps within each window while preserving that window's gap multiset changes source-held-out BA from 52.6% to 49.9% (the two task-cluster intervals overlap). These controls do not establish that precise event order or timing order is a stable cross-source A/H signal. The category-normalized A/H results are reported in Table 4.

E Conditioned association across sources

The within-source comparisons in Figure 13 and the source-held-out comparisons in Table 16 use the conditioned shallow-tree protocol. The frozen NYU test uses the matched learners and the CTH-trained thresholds described in Appendix B.

Table 16: Complete source-held-out association comparison. Training uses the other sources; B/S windows contain eight events. BA is in percent; differences and unadjusted 95% cluster intervals are in percentage points. Green/red encode improvement/decline in the point estimate; interval bounds retain the uncertainty.

Test source	Evidence	+ B/S	ΔBA [95% CI]
CTH	51.98	56.69	+4.71 [+0.69, +8.83]
KYPO	66.40	66.61	+0.21 [−1.06, +1.36]
Antieval	71.90	68.63	−3.27 [−6.28, −0.22]

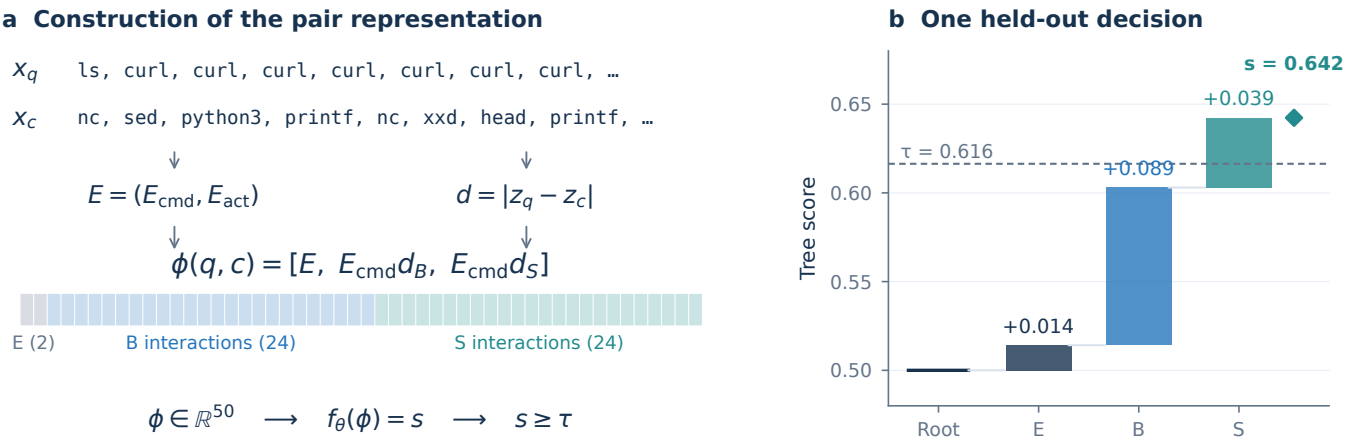


Figure 12: Earlier conditioned tree scorer and one held-out CTH `locktalk` decision. Panel a shows how two evidence scores and 48 conditioned B/S differences enter the scorer. Panel b shows contributions to a selected score among 31 corrected cross-mode same-task pairs in the purged source holdout. Each model uses its own training-selected threshold.

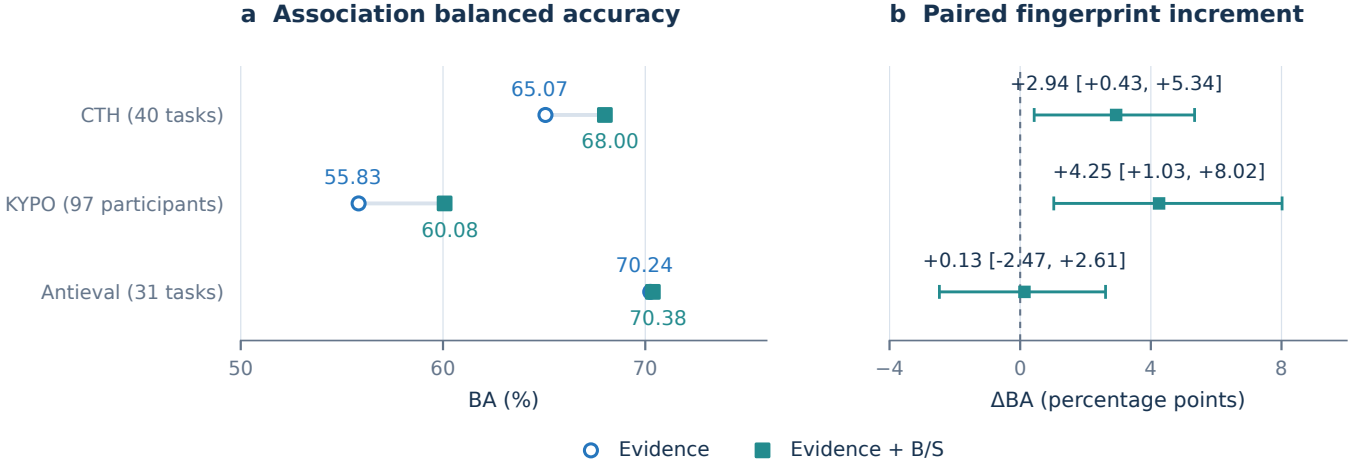


Figure 13: Earlier within-source evidence-conditioned association. Panel a compares the same shallow-tree family before and after adding B/S. Panel b shows paired BA changes and 95% intervals. CTH and Antieval resample tasks. KYPO resamples query participants with galleries fixed. The intervals are exploratory and unadjusted.